

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8873/01

Paper 1 Multiple Choice

14 Sep 2018

1 hour

Additional Materials:

Data Booklet

Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This document consists of **12** printed pages.



Innova Junior College

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 A giant molecule contains a large amount of carbon, mainly of isotopes ^{12}C and ^{13}C . It was found that the relative atomic mass of carbon in the molecule is 12.20.

What is the ratio of ^{12}C and ^{13}C ?

- A** 3:1 **B** 4:1 **C** 3:4 **D** 1:4

- 2 Which statements about relative molecular mass are correct?

- 1 It is the sum of the relative atomic masses of all the atoms within the molecule.
- 2 It is the ratio of the average mass of a molecule to the mass of a ^{12}C atom.
- 3 It is the ratio of the mass of 1 mol of molecules to the mass of 1 mol of ^1H atoms.
- 4 It is the ratio of the mass of 1 mol of molecules to the mass of 1 mol of ^{12}C atoms.

- A** 1 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 4 only

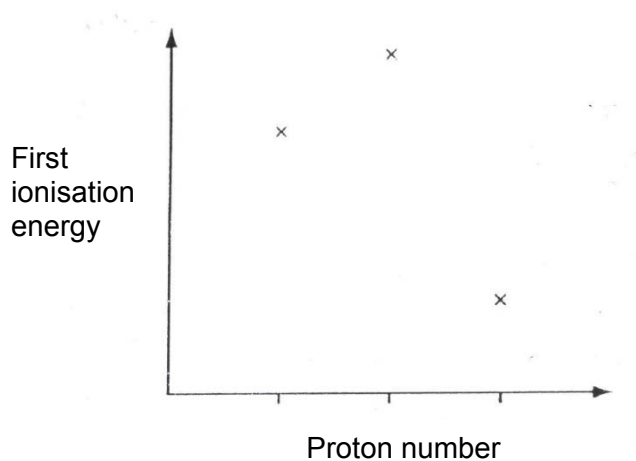
- 3 *Use of the Data Booklet is relevant to this question.*

Some isotopes are unstable and decompose naturally. In one type of decomposition, a neutron in the nucleus decomposes to form a proton, which is retained in the nucleus, and an electron, which is expelled from the atom.

Which change describes a process of this sort?

- A** $^{11}\text{C} \longrightarrow ^{12}\text{C}$
B $^{22}\text{Na} \longrightarrow ^{22}\text{Ne}$
C $^{32}\text{P} \longrightarrow ^{31}\text{P}$
D $^{40}\text{K} \longrightarrow ^{40}\text{Ca}$

- 4 Three successive elements in the Periodic Table have first ionisation energies which have the pattern shown in the diagram.



What could be the first element of this sequence?

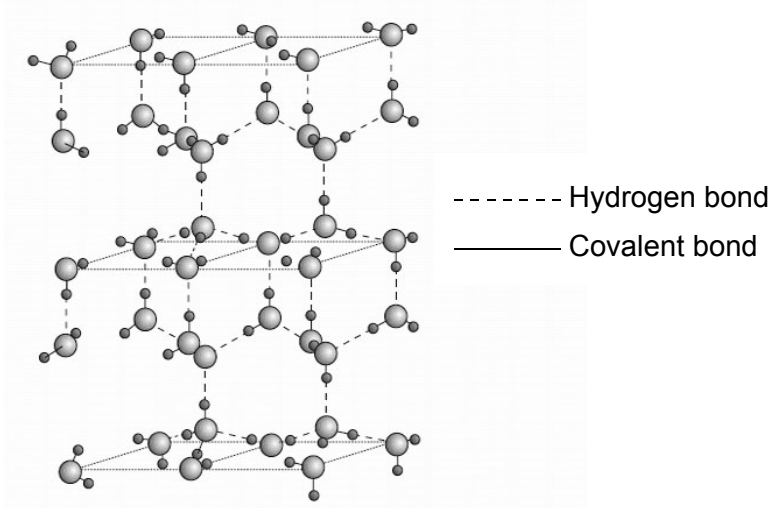
- | | | | |
|----------|---|----------|----|
| A | N | C | F |
| B | O | D | Na |

- 5 Which of the following ions contains half-filled d-orbitals?

- 1 Mn^{2+}
- 2 Cu^{+}
- 3 Fe^{2+}
- 4 Fe^{3+}

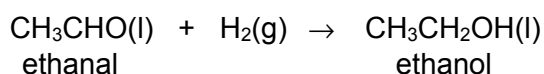
- A** 1 and 2 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 3 and 4 only

- 6** Ice is the crystalline form of water. The diagram below shows part of the structure of ice.



Which of the following statements is **not** true about ice?

- A** Ice has a lower density than water at 0 °C due to its open structure.
- B** The bond angle about oxygen in ice is 109.5°.
- C** Ice does not conduct electricity.
- D** Ice has a giant covalent structure.
- 7** Which of the following pairs of substances **does not** include a giant structure and a simple molecular structure?
- A** aluminum and silicon(IV) oxide
- B** aluminium oxide and aluminium chloride
- C** silicon and chlorine
- D** silicon and silicon(IV) chloride
- 8** The reduction of ethanal to ethanol using H₂(g) is shown in the equation below.



substance	$\Delta H_c^\theta / \text{kJ mol}^{-1}$
ethanal	-1170
hydrogen	-286
ethanol	-1370

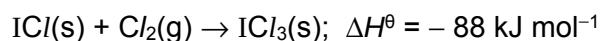
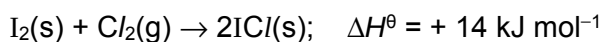
Given the standard enthalpy change of combustion in the table above, what is the enthalpy change of the reduction of ethanal to ethanol?

- A** +86 kJ mol⁻¹ **B** -86 kJ mol⁻¹ **C** -2826 kJ mol⁻¹ **D** +2826 kJ mol⁻¹

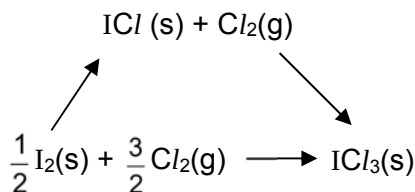
- 9 The bond dissociation energy of H–F is 565 kJ mol^{-1} . Which equation correctly describes the reaction whereby 565 kJ of energy is released?

- A $\text{HF(l)} \rightarrow \text{HF(g)}$
 B $\text{H(g)} + \text{F(g)} \rightarrow \text{HF(g)}$
 C $\text{HF(g)} \rightarrow \text{H(g)} + \text{F(g)}$
 D $\frac{1}{2} \text{H}_2\text{(g)} + \frac{1}{2} \text{F}_2\text{(g)} \rightarrow \text{HF(g)}$

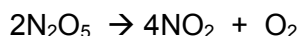
- 10 Iodine trichloride, ICl_3 , is made by reacting iodine with chlorine.



By using the above data and the following energy cycle, what is the enthalpy change of the formation of solid iodine trichloride?



- A -60 kJ mol^{-1} B -74 kJ mol^{-1} C -81 kJ mol^{-1} D -162 kJ mol^{-1}
- 11 The decomposition



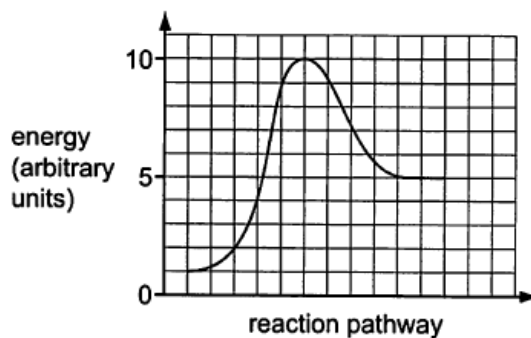
is first order with respect to N_2O_5 .

In an experiment 0.10 mol of pure N_2O_5 was put into an evacuated flask. It was found that there was 0.025 mol of N_2O_5 left 34 minutes later.

What is the time taken for the amount of NO_2 to rise from 0 mol to 0.10 mol ?

- A 17 minutes B 34 minutes C 68 minutes D 136 minutes

- 12 The diagram shows the reaction pathway diagram for an uncatalysed reaction.



The reaction is then catalysed.

What are the changes in the rate constant and the reaction pathway diagram?

	rate constant	energy profile
A	decrease	
B	decrease	
C	increase	
D	increase	

13 Which one of the following statements about the forward and backward reactions, $P + Q \rightleftharpoons R + S$, is correct when the system is at equilibrium?

- A** The ratio of the rates of the forward reaction to that of the reverse reaction equals the equilibrium constant.
- B** The rates of both the forward and the reverse reactions are equal to zero.
- C** The rates of the forward and reverse reactions are equal.
- D** The rate constant for the forward reaction equals the rate constant for the reverse reaction.

14 Each of the following equilibria is subjected to two changes carried out separately:

- (i) the pressure is reduced at constant temperature;
- (ii) the temperature is increased at constant pressure.

For which equilibrium will both of these changes result in an increase in the proportion of products?

- A** $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $\Delta H = +53 \text{ kJ mol}^{-1}$
- B** $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$; $\Delta H = -950 \text{ kJ mol}^{-1}$
- C** $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $\Delta H = -92 \text{ kJ mol}^{-1}$
- D** $N_2O_4(g) \rightleftharpoons 2NO_2(g)$; $\Delta H = +57 \text{ kJ mol}^{-1}$

15 The value of the ionic product of water, K_w , varies with temperature.

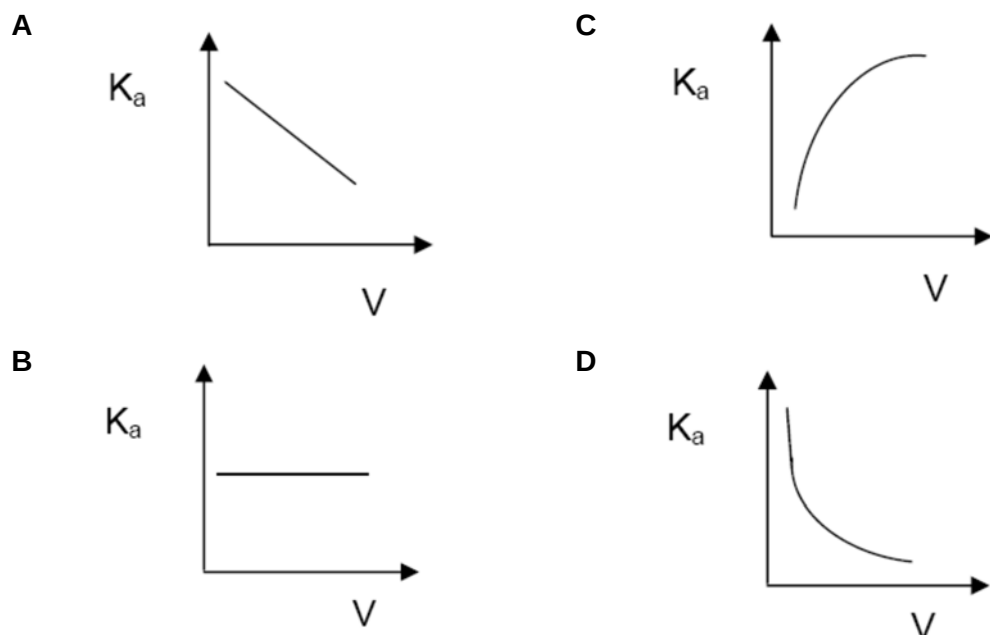
Temperature / °C	$K_w / \text{mol}^2 \text{ dm}^{-6}$
25	1.0×10^{-14}
62	1.0×10^{-13}

What can be deduced from this information?

- A** The ionic dissociation of water is an exothermic process.
- B** The association of water molecules by hydrogen bonding increases as temperature increases.
- C** The pH of pure water increases with temperature.
- D** At 62 °C, water with a pH of 6.5 is neutral.

- 16 1 mol sample of ethanoic acid is diluted at constant temperature to a volume V .

Which diagrams shows how K_a , the acid dissociation constant, varies with V ?



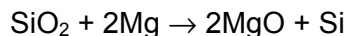
- 17 The table shows some data on two acid-base indicators.

indicator	approximate pH range	colour change	
		acid	alkali
bromocresol green	3.8 – 5.5	yellow	blue
phenol red	6.8 – 8.5	yellow	red

Which conclusion can be drawn about a solution in which bromocresol green is blue and phenol red is yellow?

- A** It is weakly acidic.
B It is neutral.
C It is weakly alkaline.
D It is strongly alkaline.
- 18 Which of the following elements forms an insoluble oxide and a chloride which is readily hydrolysed?
- A** magnesium
B phosphorus
C silicon
D sodium

- 19 In the preparation of silicon, silicon dioxide is heated with magnesium.



The product mixture contains MgO and Si only.

To separate the silicon from the product mixture, students proposed the following two possible methods.

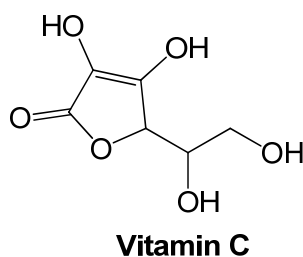
1. Shake the mixture with aqueous hydrochloric acid and filter.
2. Heat the mixture gently and collect the evaporated silicon.

Which methods would work?

- A** 1 and 2 **B** 1 only **C** 2 only **D** neither 1 or 2
- 20 Which statements concerning the third period elements (sodium to argon) and their compounds are correct?
- 1 The elements become more electronegative from sodium to chlorine.
 - 2 The melting points of the elements decrease across the period.
 - 3 The maximum oxidation state is shown by silicon.
 - 4 Aluminium oxide is the only oxide which is amphoteric.

- A** 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only

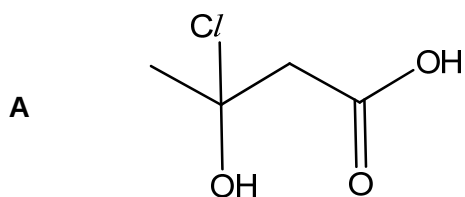
- 21 Vitamin C has the following structure.



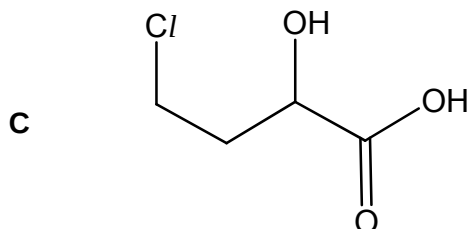
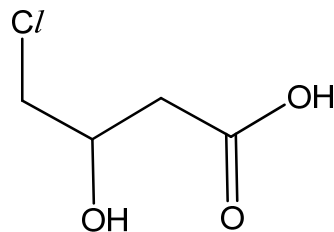
How many sp^2 and sp^3 hybridised carbon atoms does this molecule have?

	sp^2	sp^3
A	6	0
B	4	2
C	3	3
D	2	4

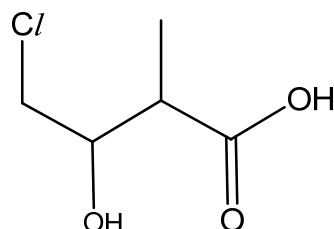
22 Which structure correctly represents 4-chloro-3-hydroxybutanoic acid?



B



D



23 As the number of carbon atoms in a homologous series of alkane molecules **increases**, for which property of the alkanes does the numerical value **decrease**?

- A density
- B enthalpy change of vaporization
- C number of isomers
- D vapour pressure

24 Compound **E**, C_4H_{10} , reacts with chlorine gas in the presence of light to form two monochlorinated alkanes, **F** and **G**, in an approximate molar ratio of 2 : 3.

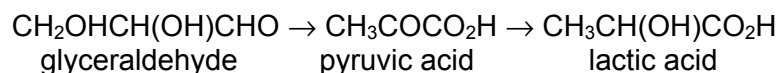
What are the structures of **E** and **F**?

	E	F
A	$CH_3CH(CH_3)CH_3$	$CH_3CH(CH_3)CH_2Cl$
B	$CH_3CH(CH_3)CH_3$	$CH_3CCl(CH_3)CH_3$
C	$CH_3CH_2CH_2CH_3$	$CH_3CH_2CHClCH_3$
D	$CH_3CH_2CH_2CH_3$	$CH_3CH_2CH_2CH_2Cl$

25 How many different alkenes, including cis-trans isomers, could be produced when 3-methylpentan-2-ol reacts with excess concentrated sulfuric acid at $170\text{ }^\circ\text{C}$?

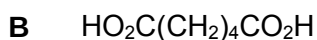
- A 2
- B 3
- C 4
- D 5

- 26 Lactic acid builds up in muscles when oxygen is in short supply. It can cause muscular pain. Part of the reaction sequence is shown.

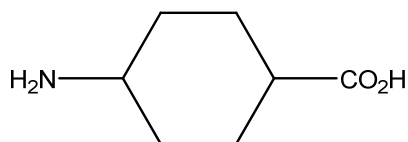


Which statements about the reaction sequence are correct?

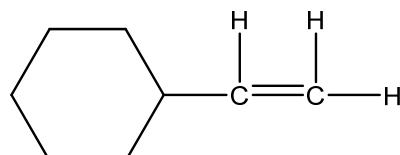
- 1 A secondary alcohol is oxidised to a ketone.
 - 2 A ketone is reduced to a secondary alcohol.
 - 3 An aldehyde is oxidised to a carboxylic acid.
 - 4 A carboxylic acid is reduced to a primary alcohol.
- A 1 and 2 only
 B 1 and 4 only
 C 1, 2 and 3 only
 D 2, 3 and 4 only
- 27 A food chemist wants to create the odour of green apples for a product. An ester with this odour has the formula $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}(\text{CH}_3)_2$. In which of the following will the substances react together to produce this ester?
- A $\text{CH}_3\text{CH}_2\text{OH}$ and $(\text{CH}_3)_2\text{CHCOOH}$
 B CH_3COOH and $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
 C $\text{CH}_3\text{CH}_2\text{COOH}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 D $\text{CH}_3\text{CH}_2\text{COOH}$ and $(\text{CH}_3)_2\text{CHOH}$
- 28 Which compound could be used by itself to form a condensation polymer?



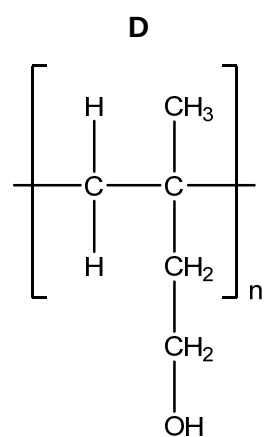
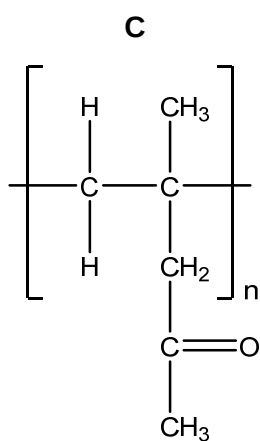
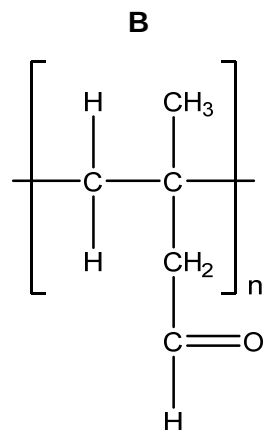
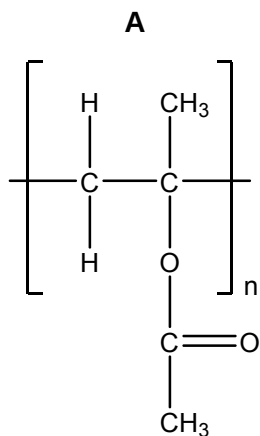
C



D



- 29 The following polymers could be used for contact lenses. Contact lenses have to absorb water so that they can fit comfortably in the eyes. Which polymer will be the most suitable to be used for contact lens?



- 30 Which are the possible ways do nanomaterials enter the human body?

- 1 skin contact
- 2 inhalation
- 3 orally taken

- A 1 only
 B 1 and 2 only
 C 2 and 3 only
 D 1, 2 and 3 only

End of Paper